

Mobile living made easy.

DOMETIC OUTDOOR



FRIDGE, FREEZER OR BOTH COMBINED



Specifications

REFRIGERATION TECHNOLOGY

Compressor

REFRIGERANT TYPE

R134a

TEMPERATURE RANGE

12 °C to 0 °C Fridge
-5 °C to -15 °C Freezer

INPUT VOLTAGE

12/24 V DC
100-240 V AC

INCLUDED

Refrigerator x 1
Drain elbow x 1



Benefits with NRX Compressor Refrigerator

- 3-in-1 cooling solution: fridge, freezer or both at the same time
- Energy-efficient high-end compressor with 3 operating modes (ECO/Performance/Quiet)
- Frameless door with dark grey front panel and full-width handle
- Beveled rear side for maximal utilisation of space on board
- Reversible door hinge – can be fitted left or right as required
- Dual function locks can also act as vents for your fridge
- Modular installation frame (standard or flush-mount)



1. Full-width door handle

Neatly integrated for easy access from every angle



2. Touch control panel

Flush-mounted inside the cabinet, with nicely integrated LED light bar



3. Removable freezer

The freezer compartment can be easily removed to make room for more fridge storage



4. Two crispers

For storing fruits and vegetables for optimal freshness



5. Well equipped interior

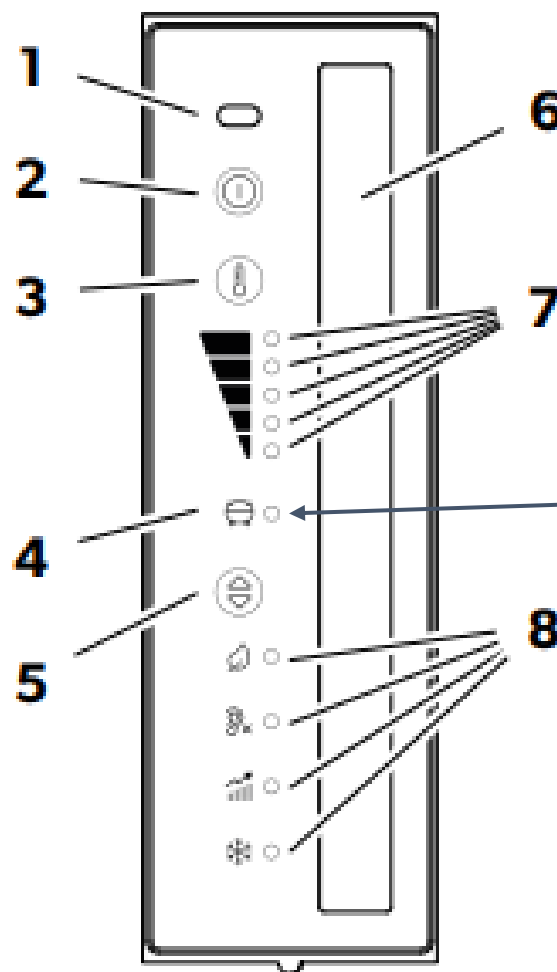
Freezer, shelves, bottle holder, door tray, crispers – it's all there



TECHNICAL DATA					
MODEL	DOMETIC NRX 50C Compressor refrigerator 50 l	DOMETIC NRX 60C Compressor refrigerator 60 l	DOMETIC NRX 80C Compressor refrigerator 80 l	DOMETIC NRX 115C Compressor refrigerator 115 l	DOMETIC NRX 130C Compressor refrigerator 130 l
SKU	9620001870	9620001872	9620001868	9620001871	9620001867
Total Volume (l)	44	55	75	113	126
Freezer Volume (l)	4.4	6.1	6.6	8.2	8.2
Freestanding Power Consumption (25°C ambient, 5°C fridge)	~ 0.77Amp/hr over 24hrs	~ 0.81Amp/hr over 24hrs	~ 1.05Amp/hr over 24hrs	~ 1.27Amp/hr over 24hrs	~ 1.29Amp/hr over 24hrs
Unique Feature	Adjustable shelves x 2 Crisper x 2 Bottle holder x 1 Upper door shelf	Adjustable shelves x 2 Crisper x 2 Bottle holder x 1 Upper door shelf	Adjustable shelves x 2 Crisper x 2 Bottle holder x 1 Upper door shelf	Adjustable shelves x 3 Crisper x 2 Bottle holder x 1 Egg shelf x 1 Upper door shelf	Adjustable shelves x 3 Crisper x 2 Bottle holder x 1 Egg shelf x 1 Upper door shelf
Dimensions W x D x H (mm)	380 x 510 x 525	448 x 537 x 525	475 x 537 x 625	525 x 568 x 745	525 x 568 x 812
Net Weight (kg)	17	18	20	26	28

Control and display elements

No in fig. 2, page 3	Explanation
1	IR door opening sensor
2	ⓘ ON/OFF button
3	🌡️ Cooling level setting button
4	Compressor and error indicator LED
5	⬆️⬆️⬆️ Operation mode button
6	Interior lighting
7	Cooling level indicator LEDs ■ Cooling Level 5 ▒ Cooling level 4 ▓ Cooling level 3 ▒ Cooling level 2 ▒ Cooling level 1
8	Operation mode indicator LEDs



Compressor and error indicator LED

The compressor and error indicator LED displays the operating status of the compressor or indicates an error:

Color	Description
White	Compressor on
Orange	Compressor off
Orange (flashing)	

Flashes	Audio?	Type	Fault	Solution/to check
Solid - on	No	Warning	Defective sensor indication	Replace sensor in Fresh food section at rear upper area
1	Yes	Warning	DC Voltage below 10 or above 31.5	Resets automatically once voltage is restored to normal range. Things to Check: • Wiring & connections • Battery charge and efficiency • Connected equipment such as solar/invertors/adaptors etc • Do they have what they need?
2	Yes	Error	External fan overcurrent	• Fan overload (Jammed or faulty)
3	Yes	Error	Compressor won't start	• Power interruptions such as on off with ignition of vehicle causing lockout • Compressor control module • Refrigerant overcharge in a new unit • Locked rotor compressor
4	Yes	Error	Compressor overload	• Shut power off for 30 minutes then restart • If large headload recently introduced, reduce the load by half and add later
5	Yes	Error	Compressor overheated	• Ventilation is adequate • External fan is running

Operation mode indicator LEDs

The operation mode indicator LEDs display the following setting options:

Symbol	Description
	ECO mode: Operation with reduced energy consumption. Cooling level 1 – 3
	QUIET mode: Operation with reduced noise level. Cooling level 1 – 4
	PERFORMANCE mode Cooling level 1 – 5
	FREEZER mode : Operation with frozen compartment removed. Cools down the complete interior of the refrigerator to – 6 °C
	WINTER mode (only NRX1115–1130): To ensure adequate cooling of the frozen compartment at low ambient temperatures.

Setting the operation modes



NOTE

When using the ECO, PERFORMANCE, FREEZER or WINTER mode the system stays in this mode until further changes.

- To set the operation mode, press the  button repeatedly until the operation mode indicator LED for the desired operation mode lights up.

The following operation modes are available:

Using the ECO mode

With ECO mode activated the refrigerator operates with reduced energy consumption at cooling levels 1 – 3.

- ✓ Operation mode indicator LED for ECO mode () and selected cooling level indicator LED light up.

Using the QUIET mode

With QUIET mode activated the refrigerator operates with reduced noise level at cooling levels 1 – 4.



NOTE

- The QUIET mode is not available if the refrigerator is running in FREEZER mode.
- When using the QUIET mode the refrigerator will stay in this mode for 8 hours and then return to the previously selected mode (ECO or PERFORMANCE mode).

- ✓ Operation mode indicator LED for QUIET mode () and selected cooling level indicator LED light up.

Using the PERFORMANCE mode

With PERFORMANCE mode activated the refrigerator operates in standard mode at cooling levels 1 – 5.

- ✓ Operation mode indicator LED for PERFORMANCE mode () and selected cooling level indicator LED light up.

Using the FREEZER mode

With FREEZER mode activated the complete interior of the refrigerator is cooled to – 6 °C.



NOTE

- Remove the frozen compartment to use FREEZER mode.
- The FREEZER mode is not available if the refrigerator is running in WINTER mode.

- ✓ Operation mode indicator LED for FREEZER mode () and all cooling level indicator LEDs light up continuously.

Using the WINTER mode for operation at low ambient temperatures (only NRX1115–1130)

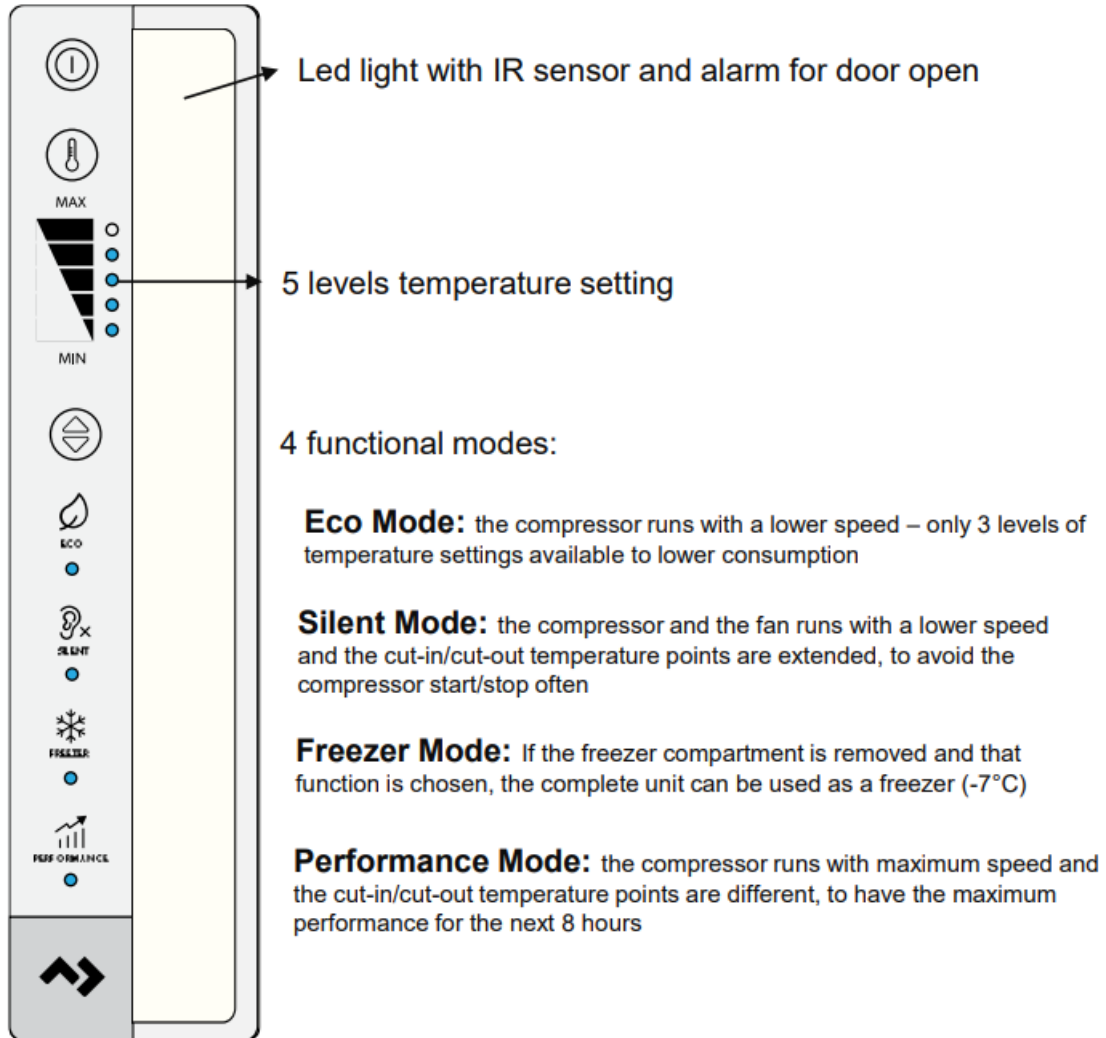


NOTE

- Cold air can impact the performance of the cooling unit. Activate the WINTER mode if you notice a loss of cooling capacity in the frozen compartment.
- Deactivate the WINTER mode as soon as the ambient temperature rises to +16 °C to avoid higher power consumption.
- The WINTER mode is not available if the refrigerator is running in FREEZER mode.

- To activate or deactivate WINTER mode proceed as shown in fig. 6, page 5.
- ✓ Operation mode indicator LED for FREEZER mode () lights up green and selected cooling level indicator LED remains on.

User Selectable modes

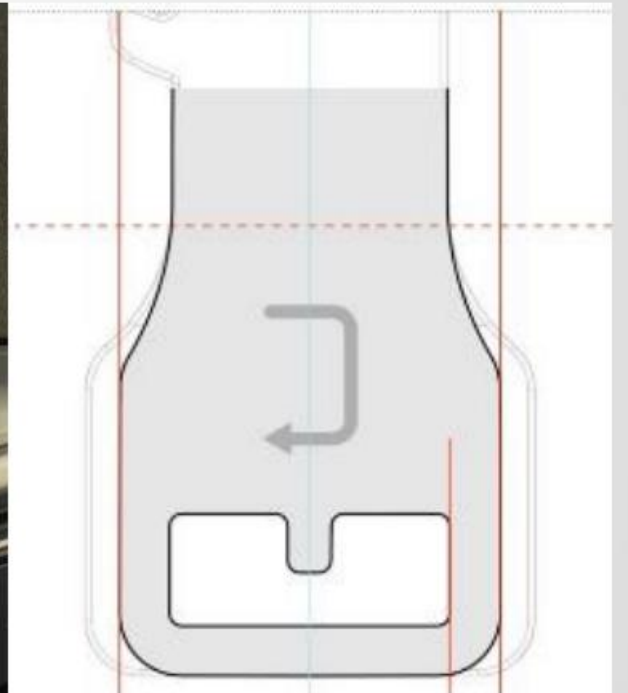


New Latch set up

(Almost as tricky as CRX!)

TOP & BOTTOM locking latches

- Introducing the New Top Locking Mechanism with Ventilation Function (Door Open).
- Clear Arrow Indicator Demonstrating the Required Movement to Open the Ventilation Position.

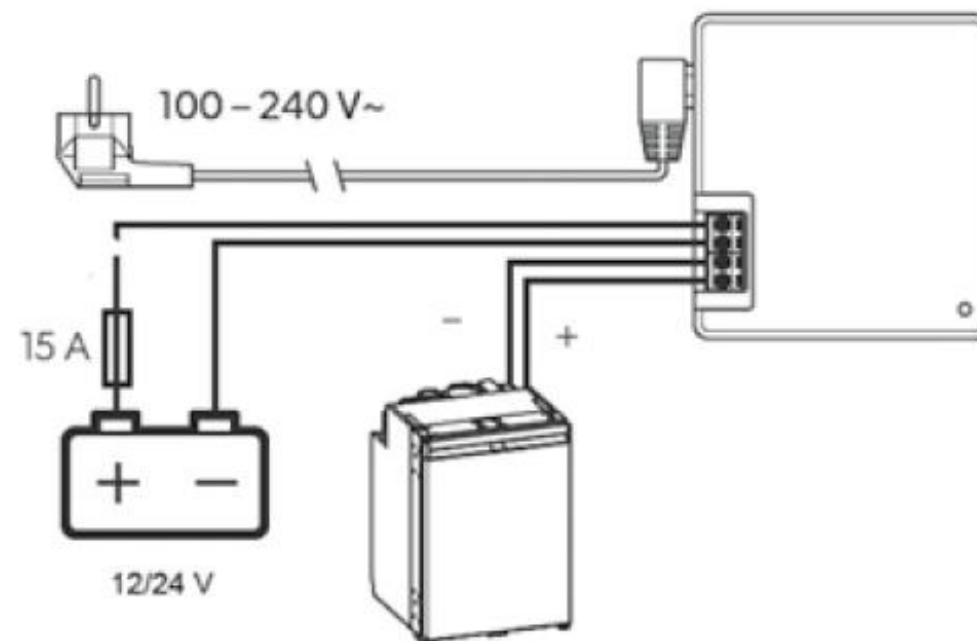


DPS-R135 - AC connection range 110–240 V, it will work with most mains systems in the world.

- Connects 24 V DC compatible refrigerators or other appliances to the mains power socket
- Compatible with all Dometic compressor refrigerators
- Easy quick connect wiring
- Automatic switching DC <> AC (AC Priority)
- Can connect to 12-volt system despite 24-volt output
- Upgrade older refrigerators with Danfoss/Secop or Waeco/Dometic compressors



- Similar to AC-MPS50 set up – 12 volt from battery runs fridge and 24 volts from adaptor runs fridge when AC is applied
- (MPS-50 was 150-Watt unit)



Comparison – NRX v CRX

	Model	50	60	80	115	130
W (Width mm)	NRX	380	448	475	525	525
	CRX	380	448	475	520	525
H (Height mm)	NRX	525	525	625	745	812
	CRX	534	525	640	745	812
D (Depth mm)	NRX	510	537	537	568	568
	CRX	500	545	528	558	620
Weight (kg)	NRX	17	18	20	26	28
	CRX	17	19	21	26	30

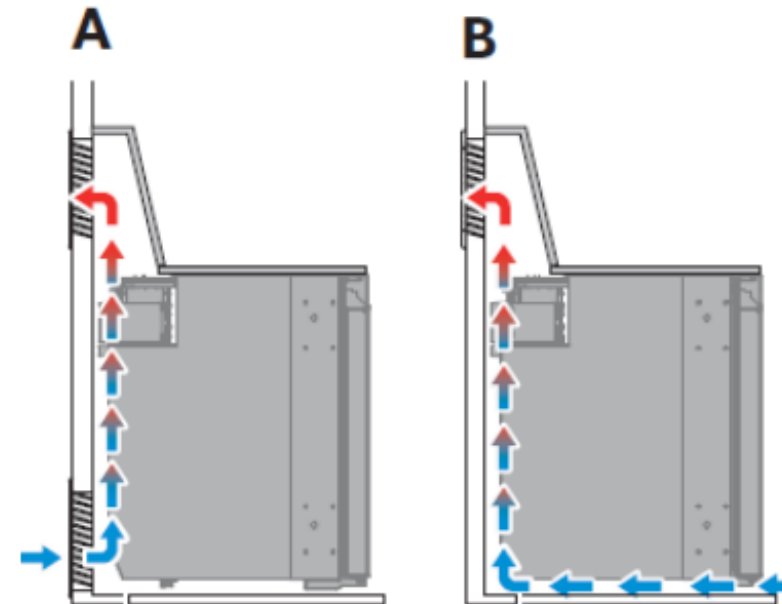
* Be careful to take note of even small differences

VENTILATION

- Insufficient air flow results in shortened lifespan and reduced cooling performance.

Preferred installation method

(External top vent does not necessarily need to be quite so high)



Having space around a refrigerator is not ventilation necessarily – say in a Ute canopy as one example. Ventilation needs air in and air out availability

Specifications– NRX Range

	NRX1050(C/S)	NRX1060(C/S)
Capacity		
– Refrigerator compartment (only NRX1050–1130[C/S])	39.6 l	48.9 l
– Frozen compartment (only NRX1050–1130[C/S])	4.4 l	6.1 l
– Total capacity	44.0 l	55.0 l
– Total capacity (frozen compartment removed)	46.0 l	58.0 l
Connection voltage	12/24 V _{DC} 100 – 240 V _{AC}	
Supply battery protection (12 V)		
– Cut-off voltage	10.4 V	
– Cut-in voltage	11.7 V	
Supply battery protection (24 V)		
– Cut-off voltage	22.8 V	
– Cut-in voltage	24.2 V	
Climate class	T	
Storage temperature range		
– Refrigerator	0 °C to +12 °C	
– Frozen compartment (only NRX1050–1130[C/S])	–15 °C to –5 °C	
– Freezer mode (only NRX1050–1130[C/S])	–6 °C	
Intended ambient temperature use	+16 °C to +43 °C	
Refrigerant	R134a	
Refrigerant quantity	26 g	30 g
CO ₂ equivalent	0.037 t	0.043 t
Global warming potential (GWP)	1430	
Noise emission (EN60704)	40 – 45dB(A)	
Dimensions H x W x D (in mm)	525 x 380 x 510	525 x 448 x 537
Net weight	17.0 kg	17.0 kg
Maximum load		
– Per door shelf	4.0 kg	5.0 kg
– Per door	8.0 kg	10.0 kg

	NRX1080(C/S)	NRX1115(C/S)	NRX1130(C/S)
Capacity			
– Refrigerator compartment (only NRX1050 – 1130[C/S])	68.4 l	104.5 l	118.0 l
– Frozen compartment (only NRX1050–1130[C/S])	6.6 l	8.2 l	8.2 l
– Total capacity	75.0 l	113.0 l	126.0 l
– Total capacity (frozen compartment removed)	78.0 l	116.0 l	130.0 l
Connection voltage	12/24 V _{DC} 100 – 240 V _{AC}		
Supply battery protection (12 V)			
– Cut-off voltage	10.4 V		
– Cut-in voltage	11.7 V		
Supply battery protection (24 V)			
– Cut-off voltage	22.8 V		
– Cut-in voltage	24.2 V		
Climate class	T		
Storage temperature range			
– Refrigerator	0 °C to +12 °C		
– Frozen compartment (only NRX1050–1130[C/S])	–15 °C to –5 °C		
– Freezer mode (only NRX1050–1130[C/S])	–6 °C		
Intended ambient temperature use	+16 °C to +43 °C		
Refrigerant	R134a		
Refrigerant quantity	32 g	40 g	40 g
CO ₂ equivalent	0.046 t	0.057 t	0.057 t
Global warming potential (GWP)	1430		
Noise emission (EN60704)	40 – 45dB(A)		
Dimensions H x W x D	625 x 475 x 537 mm	745 x 525 x 568 mm	812 x 525 x 568 mm
Net weight	20.0 kg	25.0 kg	26.0 kg
Maximum load			
– Per door shelf	5.0 kg	6.0 kg	6.0 kg
– Per door	10.0 kg	18.0 kg	18.0 kg

DC wiring to Refrigerator

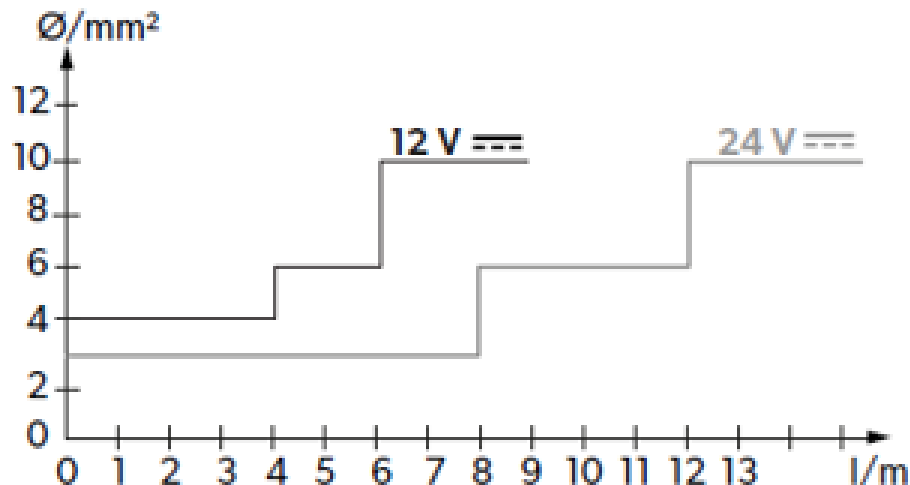
Wiring needs to be sufficiently heavy to carry the load. The thinner the incoming wiring to the refrigerator, the greater the voltage loss.

The longer the run of wiring from the battery to the refrigerator, the greater the voltage loss. All joins/connections along the run are potential voltage loss areas.

Under load, that is as the compressor starts, the voltage loss is increased substantially; without load, that is when the compressor is not running, you will be able to measure what appears to be suitable voltage to enable the refrigerator to run.

The refrigerator compressor can try to start, which causes a voltage loss due to inadequate wiring and the compressor shuts down and a one flash error appears. Immediately the compressor stops, the load is no longer there, so the voltage once again shows higher, and the compressor tries to start again and once again shuts down as the voltage drops. (Over and over)

Don't be fooled, always measure the available voltage under load conditions. Below table is a good guide for wiring to compressor refrigerators.



Solar and battery storage

When fitting a solar electrical system, it is important to know what is the likely power drain from the battery storage every day; add up all of the consumers and a good rule of thumb is to try to have around 2/3 in storage.

Example: You expect to consume 100 amphotours each day in total, good practice would be to have 300 Amphotours of battery storage capacity; this helps with cloudy days limiting solar input. It is always better to have more than you need.

What size solar?

With the “electrification” of the industry, much bigger and more elaborate systems are being fitted.

For a basic system, a rough guide/formula to refer could be:

Solar Panel watts, divided by DC voltage, multiplied by 5 sunlight hours per day. SO:

120-Watt panel on a 12 volts system equals 10.

10 multiplied by 5 good sunlight hours equals 50.

Potential provision around 50 Amphotours per day.

So, if you needed 100 Amphotours per day, you would need at least 2 X 120-Watt panels, and it would be better to have more if you have the real estate on the roof of the RV.

Personal opinion, If I want 100 amphotours per day, I am going to fit 300 watts of solar panel & 300 Amphotours of battery. I would go Lithium also as it charges more quickly and holds a higher voltage for longer than AGM/Deep cycle batteries.

Make sure the charger and regulator are MPPT suitable for Lithium.